

Introductory Project (1DT308)

Visualization

Arslan Musaddiq¹

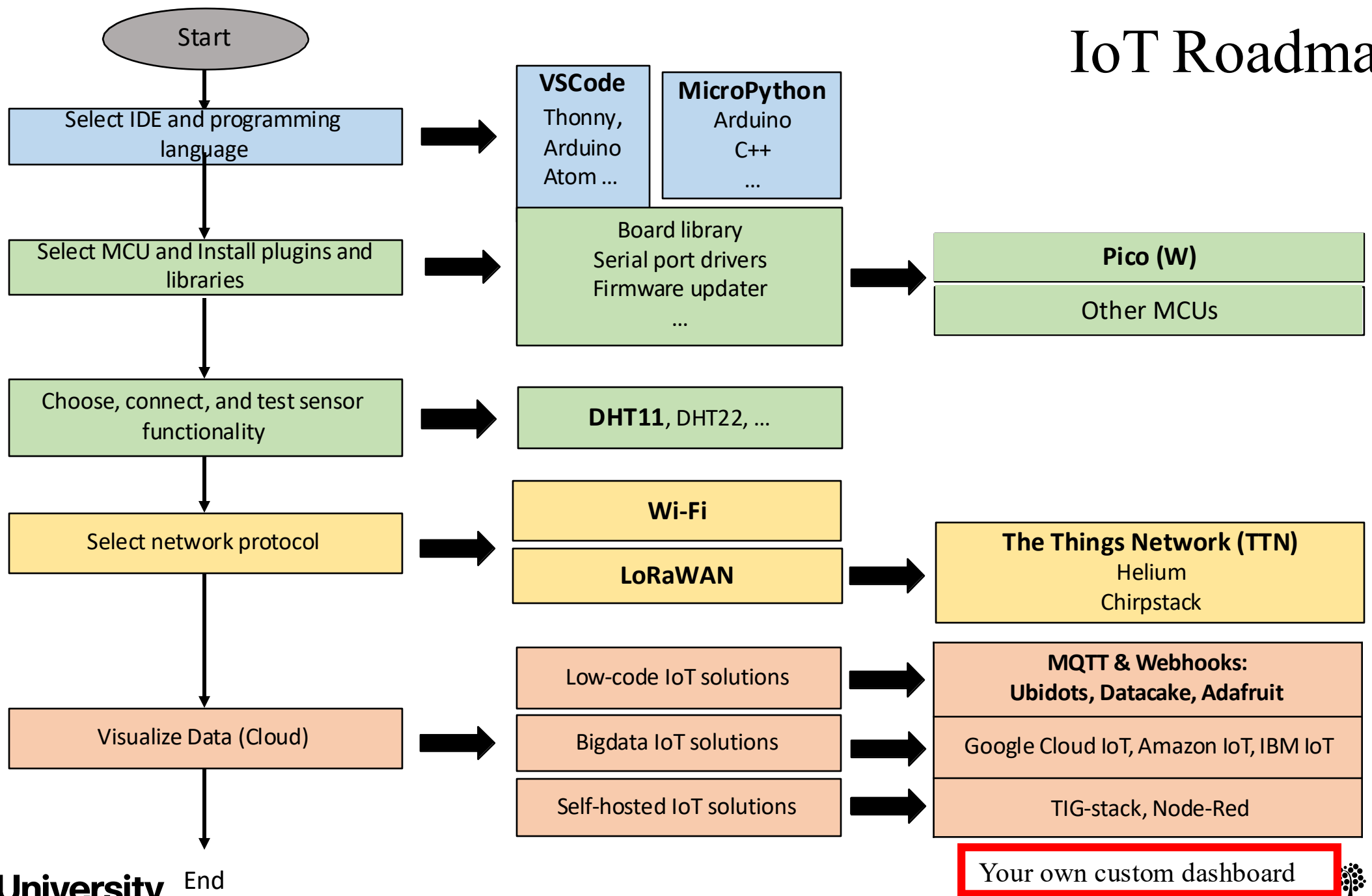
Faculty of Technology

Department of Computer science and Media Technology

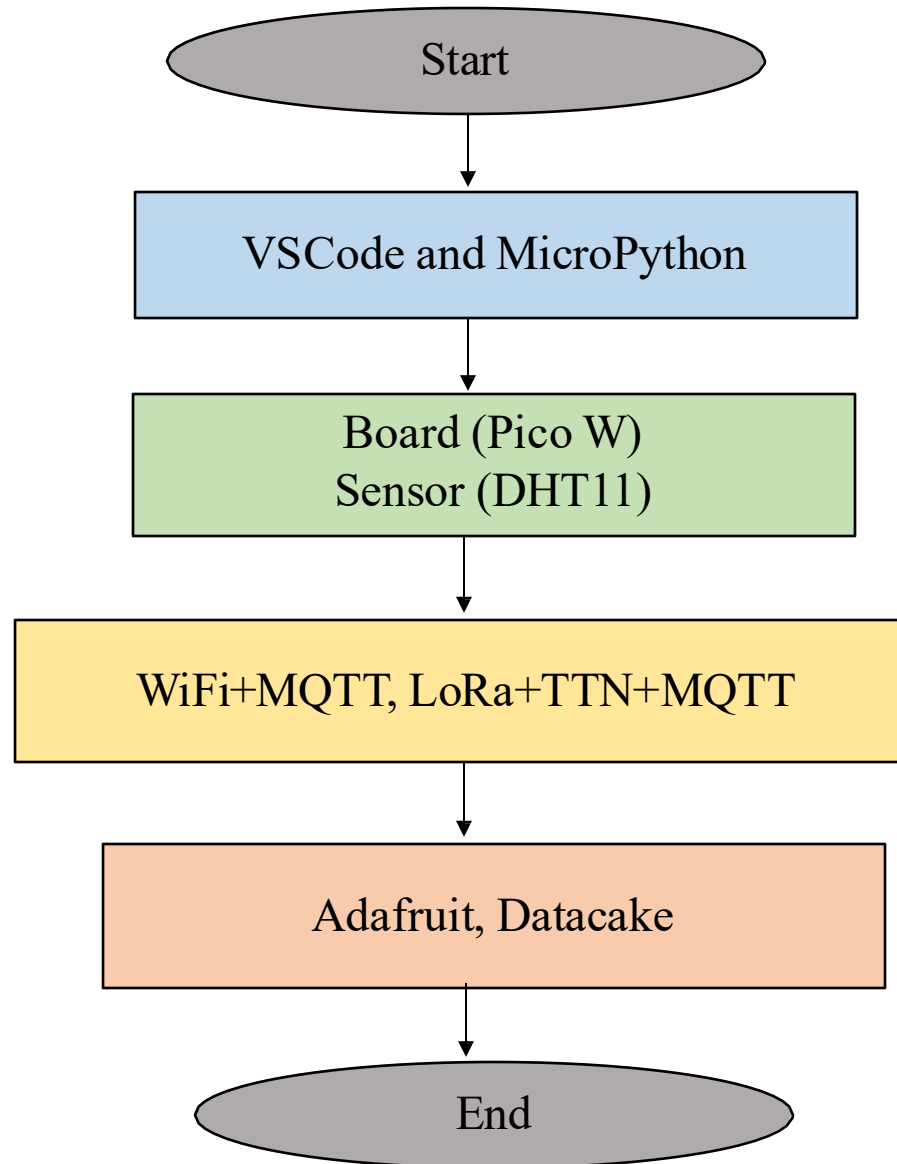
Linnaeus University, Sweden.



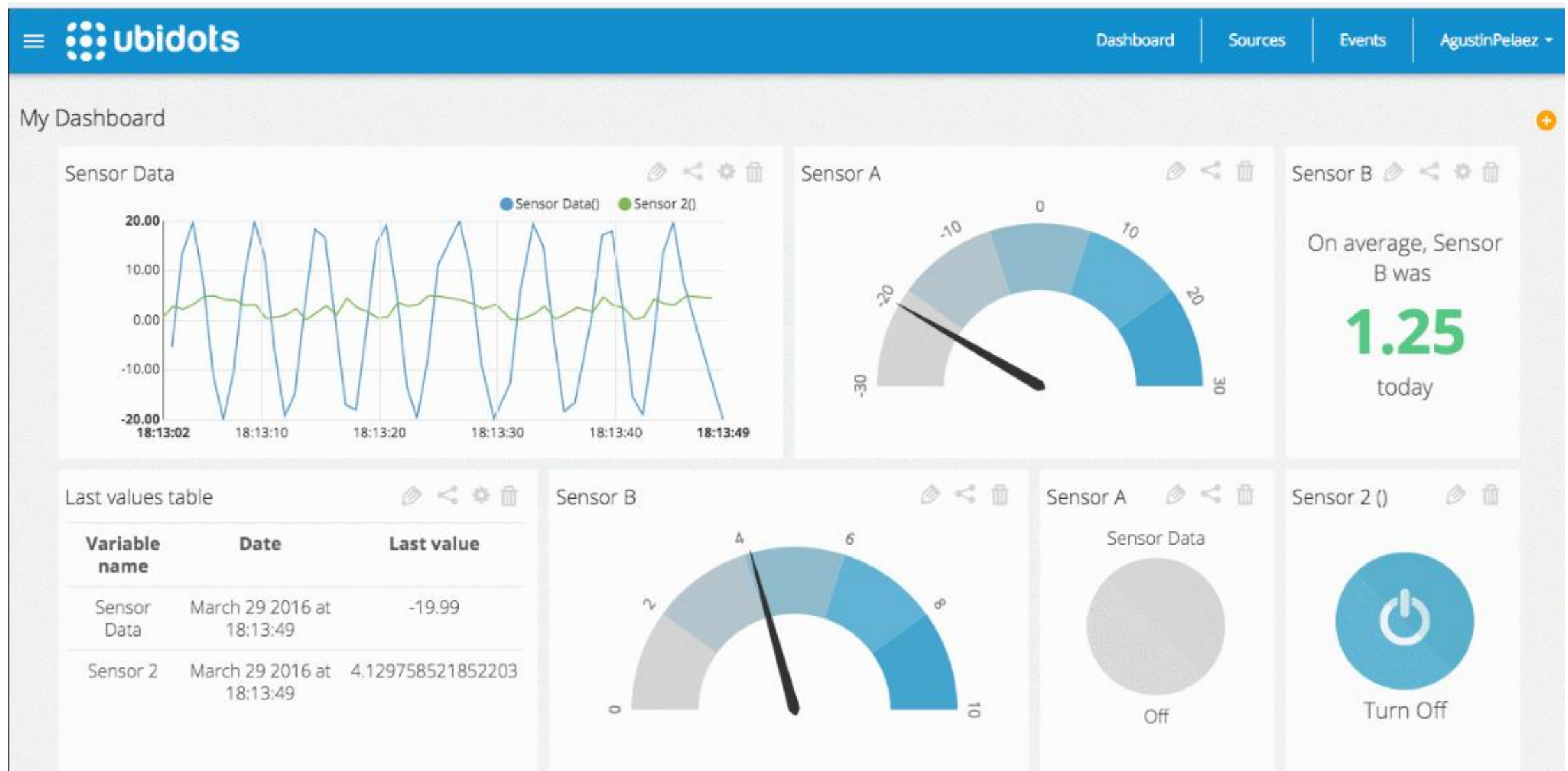
IoT Roadmap

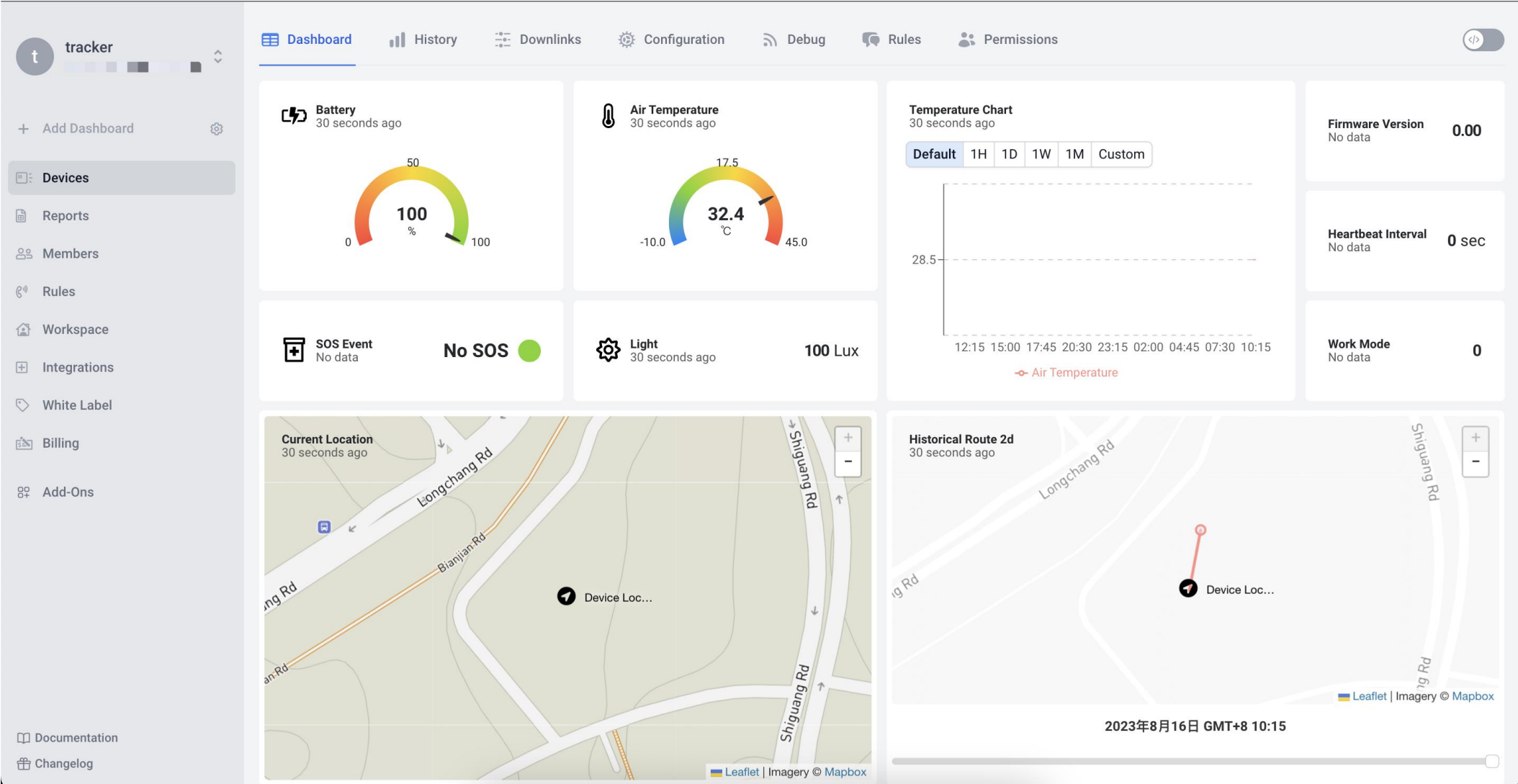


A Possible Path to Follow



Ubidots





The Things Network (TTN)

 THE THINGS STACK
SANDBOX

Home

ApplicationsGateways

Search

K

Dashboard

Organizations

Notifications

User settings

Top entities

real-iot-lab

real-iot-lab/sensitive-sensor-kitchen

Office sensors in Kalmar Magna

MeteoWind IoT Pro

Inu-kalmarslott

msdessi

Show more

Home > Dashboard

+ Add

☆

THE THINGS NETWORK

Create application

Add end device

Create organization

Add API key

Register gateway

☆ Top entities

AllApplicationsGatewaysEnd devices

TYPE	NAME	STATUS / LAST SEEN
	real-iot-lab real-iot-lab	1 min. ago
	real-iot-lab/sensitive-sensor-kitchen	1 min. ago
	Office sensors in Kalmar Magna office-kalmar	52 sec. ago
	MeteoWind IoT Pro office-kalmar/eui-0004a30b00f79436	2 min. ago
	Inu-kalmarslott	58 min. ago

📧 Notifications 9

View all →

MESSAGE	TIME
New API key created for application A new API key has just been created for your application real-iot-lab on...	Aug 22, 2024
Collaborator of application added or updated A collaborator of your application real-iot-lab on your network has been...	Aug 22, 2024
New API key created for gateway A new API key has just been created for your gateway eui-ac1f09fffe06...	Feb 15, 2024
API key of application has been changed An API key of your application "office-kalmar" on your network has bee...	Feb 13, 2024
New API key created for application A new API key has just been created for your application office-kalmar ...	Feb 13, 2024

📰 Latest news

Blog →

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March 20, 2025

The Things Conference 2025 Expands Collaboration with EDGE AI FOUNDATION to Drive Innovation in Low-Power IoT and Edge AI
March 03, 2025

The Things Conference 2025: More Edge, More IoT, More Access!
February 27, 2025

📖 Documentation

Getting started

→

End devices

→

Gateways

→

Integrations

→

The Things Stack

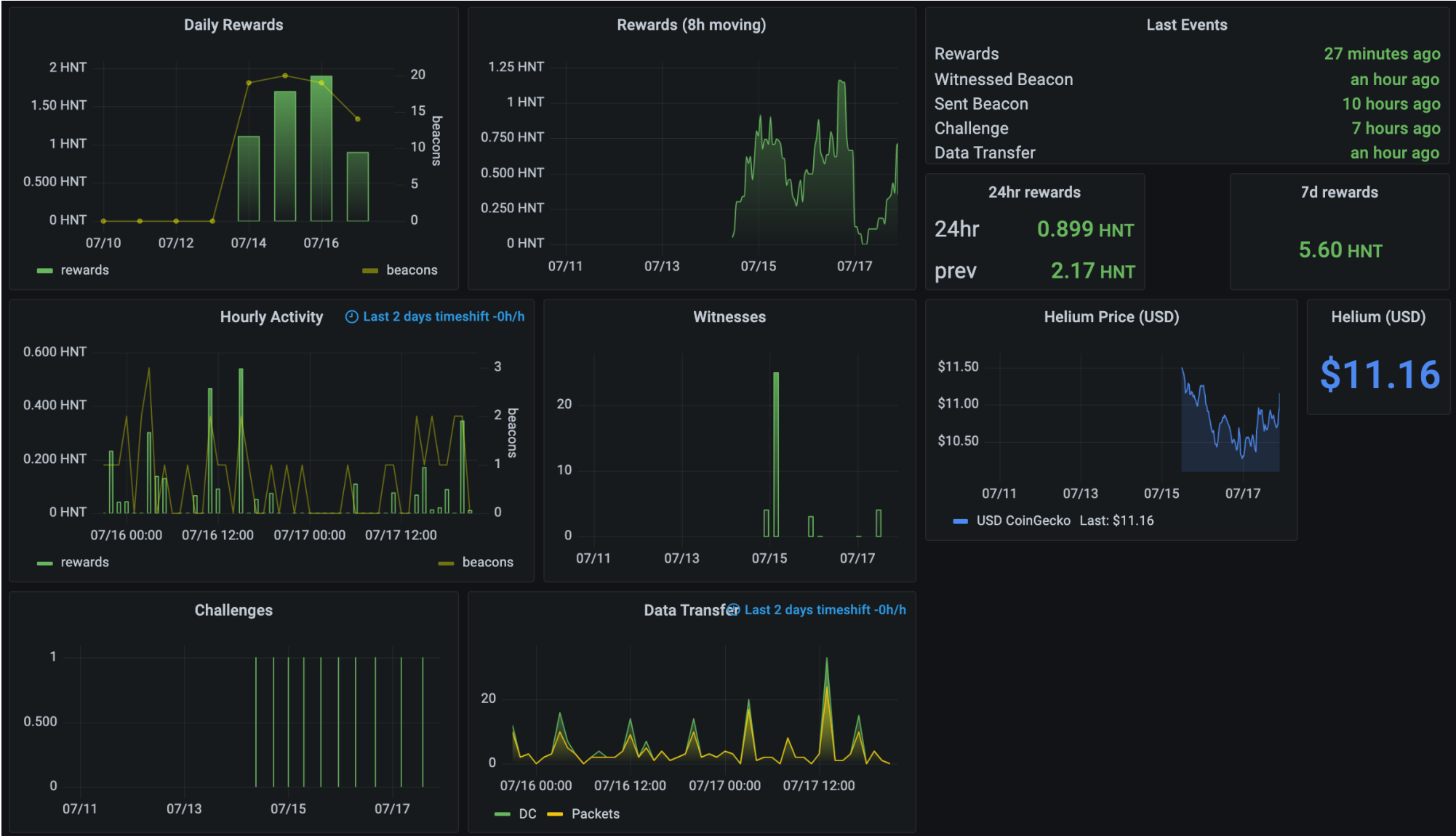
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Reference

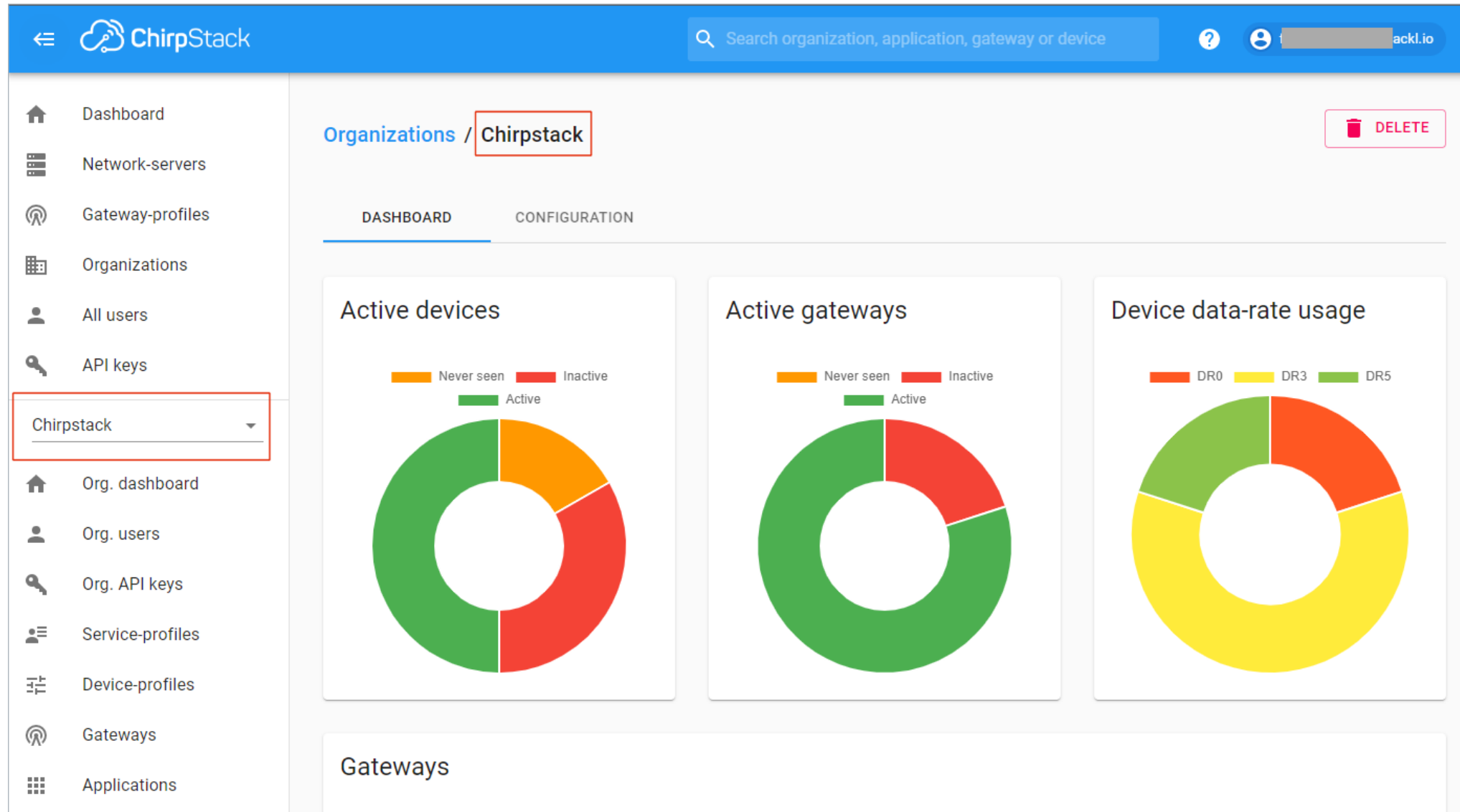
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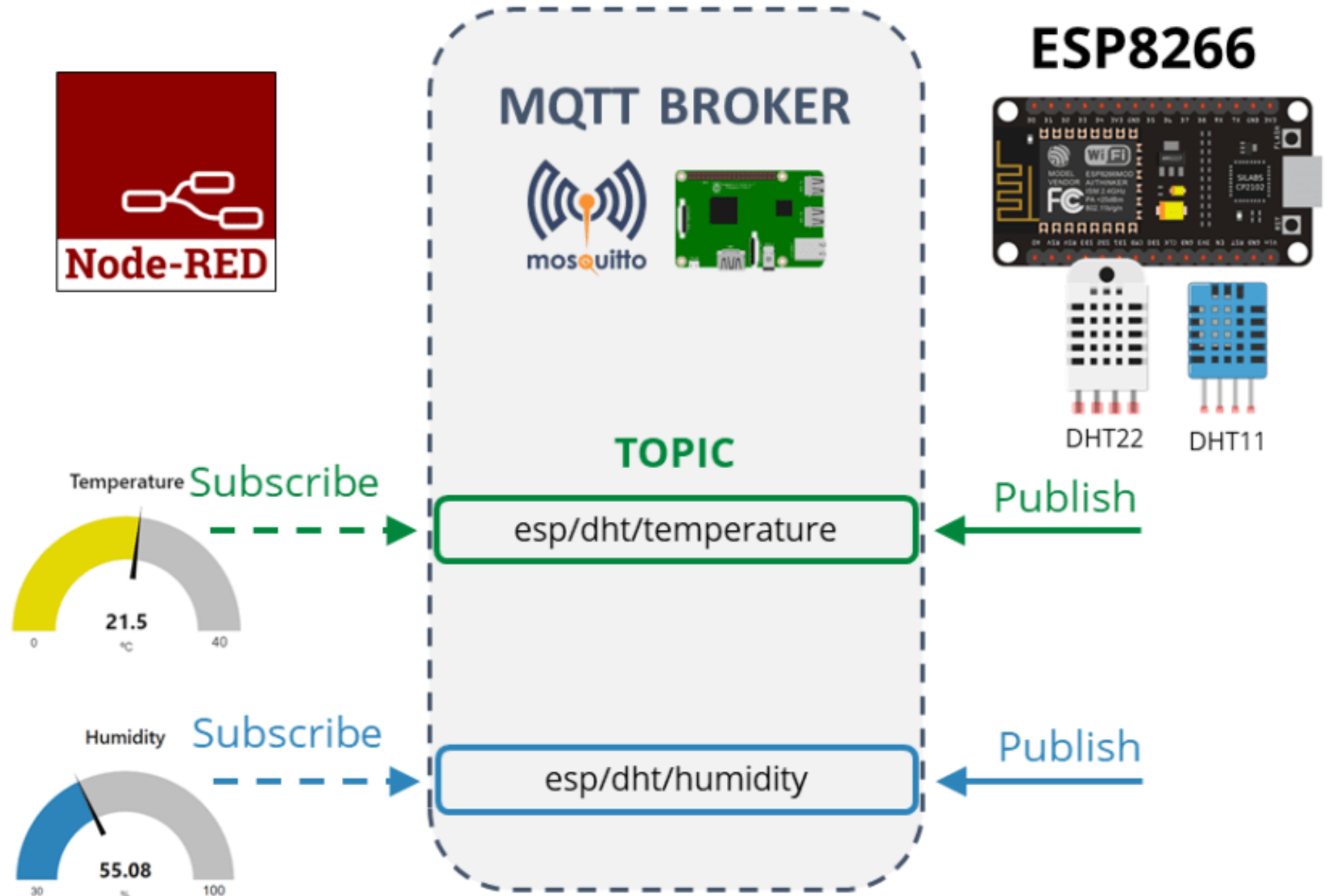
Helium Network



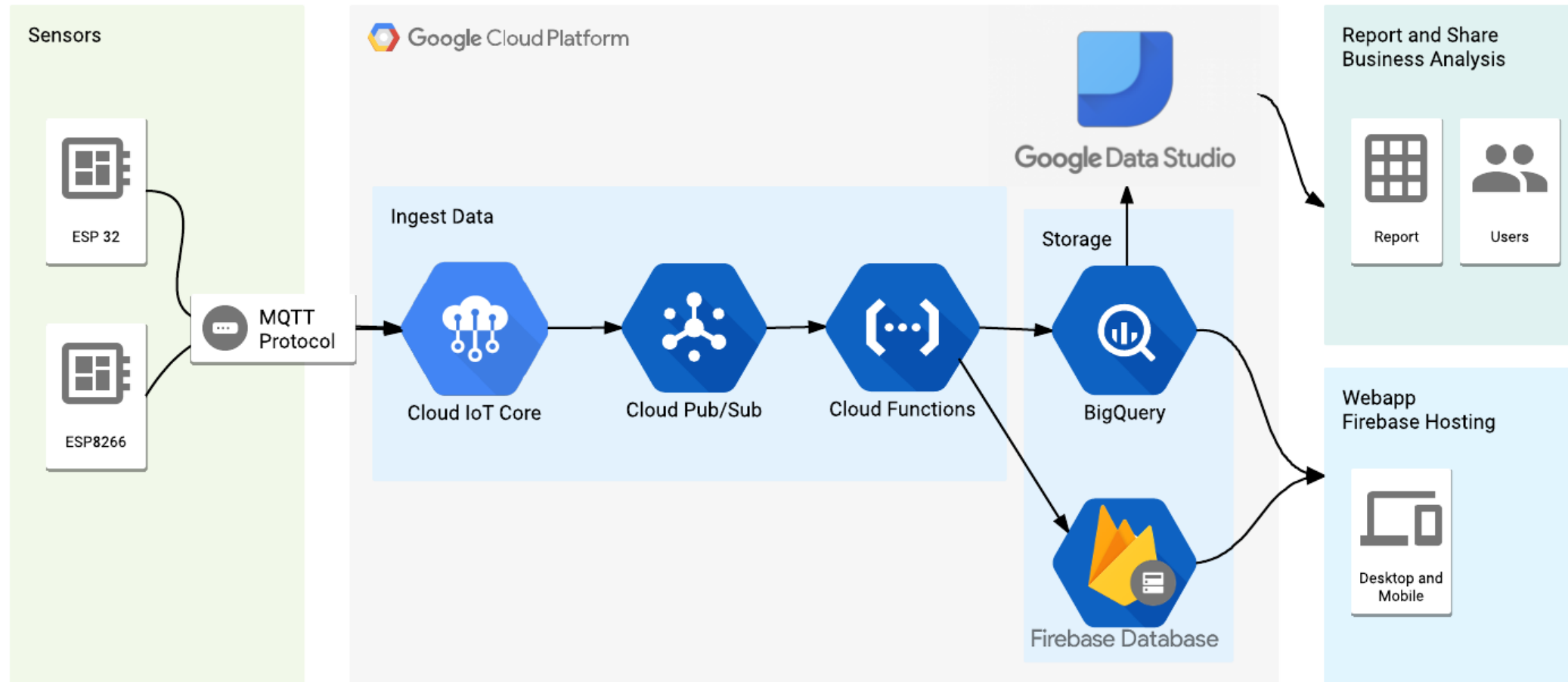
Chripstack



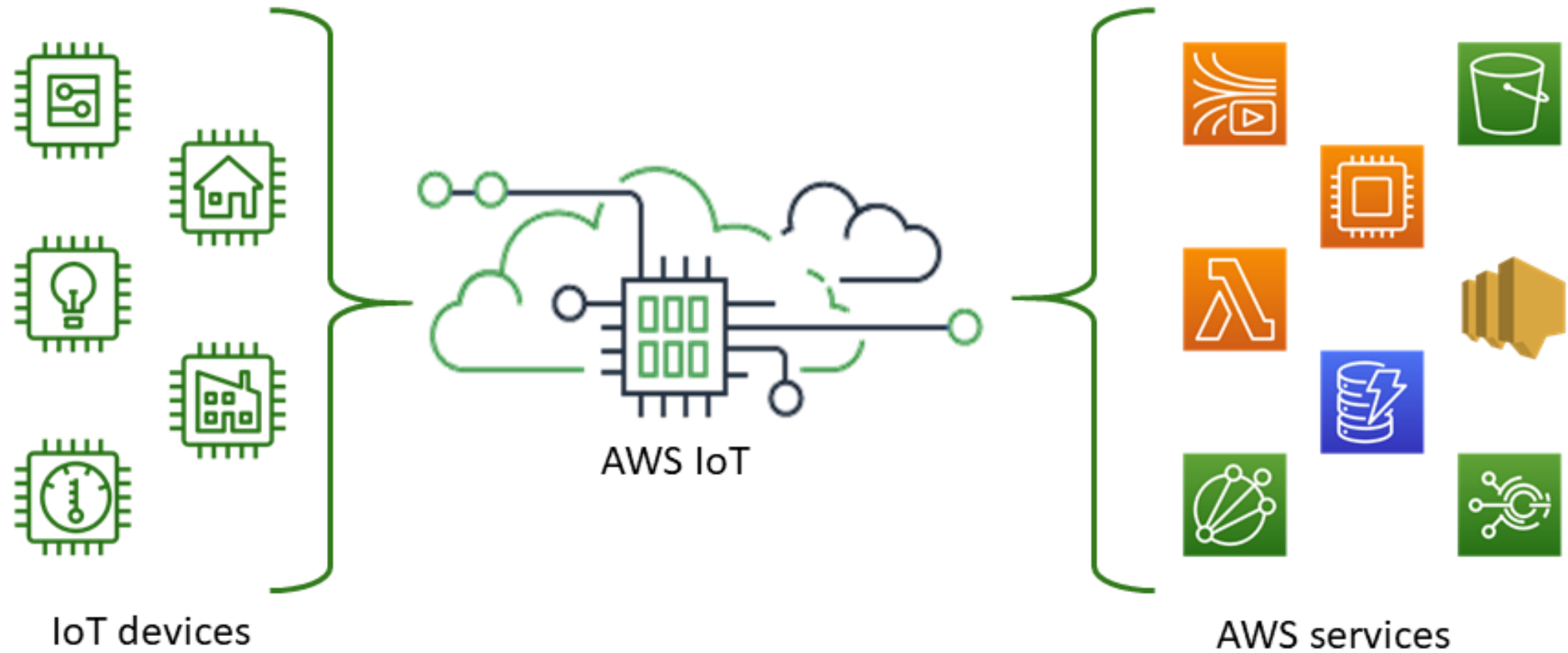
Node-Red



Google Cloud IoT



AWS IoT



Your Own Solution

Pico W → publishes → **Mosquitto Broker** → received by → **Python Subscriber** →
saves to → **CSV File** → displayed by → **Flask Web Dashboard**



Create Subscriber

What is a Subscriber?

- A subscriber **listens** for messages from a broker
- **Subscribes** to specific topics (like "test/temperature")
- **Receives and processes** incoming data

Step 1: Import Required Libraries

You need

- **csv** for saving data
- **os** to check files
- **datetime** for timestamps
- **paho.mqtt.client** for MQTT connection

Step 2: MQTT connection

You need

- CSV file name
- Broker IP address and port
- Topics to subscribe

Step 3: Setup MQTT Client with Callbacks

You need to

- Define what happens when client **connects** (subscribe to topics)
- Define what happens when a **message** arrives (save data into CSV)
- Setup client, assign callbacks, and connect to the broker



Setup flow to create Python MQTT Subscriber

MQTT Subscriber Python Script

1. Import Libraries

Imports all necessary libraries. (such as csv, os, datetime, paho.mqtt.client)

2. Check/Create CSV File

If "file.csv" doesn't exist, create it with headers: (timestamp, sensor_type, value)

3. Setup MQTT Client

Create an MQTT client object (client = mqtt.Client()).

- what to do when **connected** (subscribe to topics)

- what to do when **a message arrives** (save data)

4. Connect to MQTT Broker

Connect to broker using its IP address (client.connect()).

5. Connect to MQTT Broker

Connect to broker using its IP address (client.connect()).

6. Listen Forever (loop_forever)

Keep the connection alive.

Wait for incoming messages.

7. When Connected (on_connect)

Subscribe to test/temperature and test/humidity topics.

8. When Message Arrives (on_message)

Read the topic and value.

Find whether it's **temperature** or **humidity**.

Add timestamp.

Save everything (timestamp, sensor type, value) into the CSV file.

9. Repeat and Keep Saving

Every new sensor value gets saved into the CSV automatically.



Flask Web Application

What is Flask:

- Flask is a tool that helps you build web applications using Python

Step 1: Import Required Libraries

You need

- **Flask:** it's a main class
- **render_template:** to render HTML templates
- **Jsonify:** to convert data into JSON format

Step 2: Initializing the Flask Application

For example,

- **Flask(__name__)**
- Store the name of csv file.

Step 3: Define Main (/) and Data Route (/data)

You need to

- Define the main route of your web app (i.e., when you visit localhost:5000 in your browser) to load and display index.html.
- Defines another route (/data). When you visit localhost:5000/data, it triggers the data() function. It will read the sensor data from the CVS file and sends it to the browsers in JSON format.



HTML Structure and Basic Setup

- You can create a simple HTML page, that will use Flask to serve data.
- Flask is a tool that helps you build web applications using Python

Step 1: Create basic structure

You need

- Title
- Script: Use Chart.js to create charts
- Style: for layout and colors
- Create body section.

Step 2: JavaScript

You need

- A function to convert timestamp into readable formats.
- A function to fetch data served by Flask.
- Function to initialize the data using Chart.js.
- Also add a function to update the chart, let say every 5 seconds.

Step 3: Flask integration

- The /data route defined in the Flask app to send sensor data in JSON format to the front-end (HTML page).



LAB 3: Build Your Own IoT Dashboard and Logger

Go to the following link (Lab 3) for complete and detailed instructions:

Lab 3 - Build Your Own IoT Dashboard and Logger

https://gitlab.lnu.se/1dt308/content/-/blob/master/labs/3_Visualization.md?ref_type=heads

Take your time going through it, and feel free to ask if anything's unclear.





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